PROJECT REPORT

PWM Generation Using 555 Timer IC (Proteus Simulation)

1. Introduction

Pulse Width Modulation (PWM) is a widely used technique in electronic circuits to control power efficiently.
It works by varying the ON and OFF time of a digital signal. In this project, PWM is generated using a 555 Timer IC
and simulated using Proteus software. The simulation provides a clear understanding of PWM behavior, waveform
generation, and duty cycle variation without requiring physical components.

2. Objective

To design and simulate a PWM generator circuit using a 555 Timer IC in Proteus, observe waveform output,
and study duty cycle variation with a potentiometer.

3. Theory of PWM

PWM adjusts the average voltage by changing the pulse width of a periodic signal.
Duty Cycle (%) = (TON / (TON + TOFF)) × 100
Applications:
• Motor Speed Control
• LED Brightness Control
• Power Electronics
• Signal Modulation
• Robotics

4. 555 Timer IC Overview

555 Timer consists of:
• Two Comparators
• Flip-Flop
• Discharge Transistor
• Voltage Divider
It operates in:
1. Astable Mode
2. Monostable Mode
3. Bistable Mode
For this project, astable mode with charge/discharge control is used.

5. Software Used — Proteus ISIS

Proteus is a circuit simulation tool used for:
• Circuit design

• Waveform analysis
• Debugging
• Parameter variation
It allows real■time visualization of PWM behavior.

6. Components Used

• 555 Timer IC
• 100kΩ Potentiometer
• 10kΩ Resistor
• 10nF Capacitor
• 1N4148 Diodes × 2
• DC Power Supply
• Oscilloscope (virtual)
• Connecting Wires

7. Circuit Diagram Description

The PWM circuit uses:
• Pin2 & Pin6 shorted
• 10nF capacitor to ground
• Diodes separating charge/discharge paths
• Potentiometer adjusting duty cycle
• Output at Pin3
• Pin5 connected to 0.01μF capacitor
This ensures variable duty cycle with nearly constant frequency.

8. Simulation Procedure

1. Open Proteus ISIS.
2. Place 555 Timer, potentiometer, resistors, diodes, capacitor, oscilloscope.
3. Wire the circuit as per design.
4. Connect oscilloscope to Pin 3.
5. Apply 5V supply.
6. Run simulation.
7. Adjust potentiometer to observe PWM changes.
8. Record readings.

9. Observations

Pot Position	Frequency (kHz)	Duty Cycle (%)
Minimum	1.25	8%
25%	1.32	28%
50%	1.40	50%
75%	1.35	74%
Maximum	1.28	92%

Duty cycle changed significantly while frequency remained stable.

10. Calculations

Formulas:

tON = 0.693 × Rcharge × C
tOFF = 0.693 × Rdischarge × C
Frequency = 1 / (tON + tOFF)
Duty Cycle = tON / (tON + tOFF)
Values match simulation output closely.

11. Advantages

• No physical hardware required
• Easy debugging
• Real■time waveform view
• Safe for beginners
• Fast and accurate testing

12. Applications

• Motor control circuits
• LED brightness controllers
• Power electronics trainers
• Robotics PWM drivers
• Signal modulation systems

13. Results

PWM waveform successfully generated in Proteus. Duty cycle adjustable from 8% to 92%. Frequency remained stable around 1–1.5 kHz. The simulation accurately demonstrated the PWM operation of the 555 Timer IC.

14. Conclusion

The Proteus-based PWM generator using 555 Timer IC performed successfully. Duty cycle variation was smooth and matched theoretical expectations. The simulation effectively demonstrates PWM functionality without needing physical components.

15. References

• NE555 Timer Datasheet
• Proteus ISIS User Guide
• Electronics Tutorials
• YouTube Educational Videos